

MEMORY REFERENCE INSTRUCTIONS

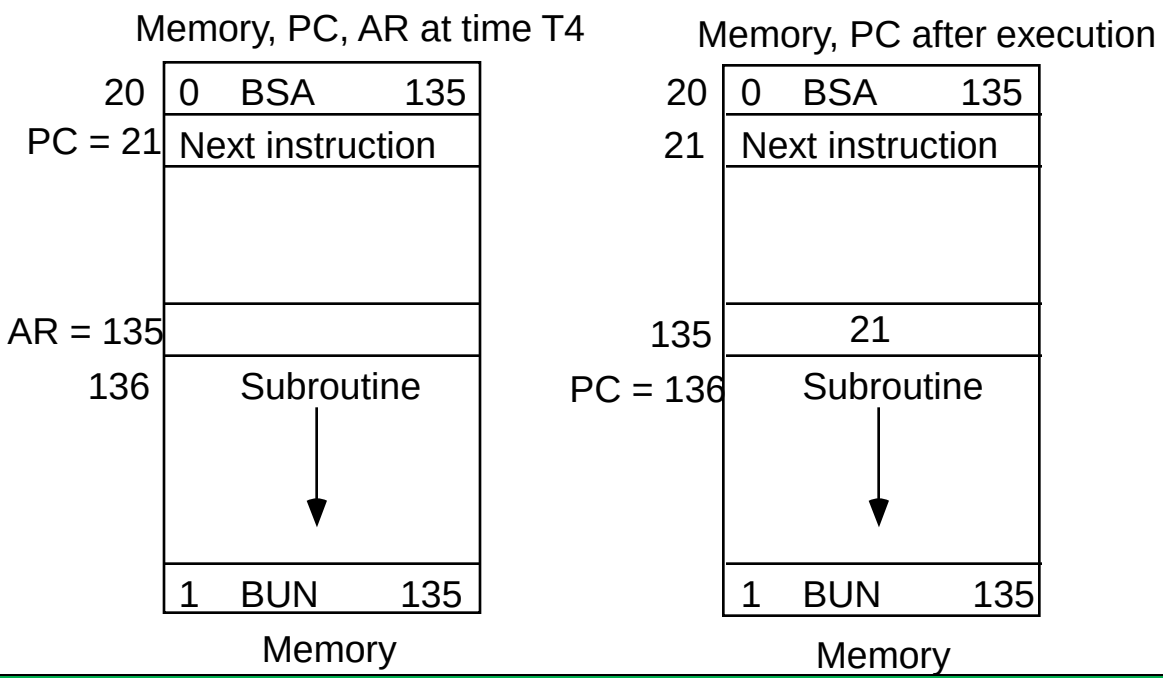
- LDA: Load to AC
- $D_2T_4:$

$DR \leftarrow M[AR]$
- $D_2T_5:$

$AC \leftarrow DR, SC \leftarrow 0$
- STA: Store AC
- $D_3T_4:$

$M[AR] \leftarrow AC, SC \leftarrow 0$
- BUN: Branch Unconditionally
- $D_4T_4:$

$PC \leftarrow AR, SC \leftarrow 0$
- BSA: Branch and Save Return Address
- $M[AR] \leftarrow PC, PC \leftarrow AR + 1$



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BSA:

D_5T_4 : $M[AR] \leftarrow PC$, $AR \leftarrow AR + 1$

D_5T_5 : $PC \leftarrow AR$, $SC \leftarrow 0$

ISZ: Increment and Skip-if-Zero

D_6T_4 : $DR \leftarrow M[AR]$

D_6T_5 : $DR \leftarrow DR + 1$

D_6T_6 : $M[AR] \leftarrow DR$, if $(DR = 0)$ then $(PC \leftarrow PC + 1)$, $SC \leftarrow 0$

